U-C-00211D (DSA-DM)
September 6, 1974
SUPERSEDING
Int. Fed. Spec.
U-C-00211C (DSA-DM)
November 17, 1972, and
Interim Revision of
Fed. Spec. U-C-211a
December 13, 1956

INTERIM FEDERAL SPECIFICATION

CEMENT, ZINC PHOSPHATE, DENTAL

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal specification U-C-211a.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type (fine grain) of zinc phosphate cement.

1.2 Classification. Zinc phosphate cement shall be furnished in the following shades as specified (see 6.1).

Shade No. 1	Light gray
Shade No. 2	Light yellow

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

FSC 6520

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SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.

(Activities, outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.)

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

SPECIFICATION

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof Or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN CHEMICAL SOCIETY (ACS) PUBLICATIONS

Reagent Chemicals, ACS Specifications

(Application for copies should be addressed to the American Chemical Society Publications, 1155 Sixteenth Street, N.W., Washington, D.C. 20036.

AMERICAN DENTAL ASSOCIATION (ADA)

ADA Specification No. 8 for Dental Zinc Phosphate Cement

(Application for copies should be addressed to the Council on Dental Materials and Devices, American Dental Association, 211 East Chicago Avenue, Chicago, Illinois 60611.)

3. REQUIREMENTS

3.1 Material. Zinc phosphate cement shall consist of a powder and a liquid which, when mixed together according to manufacturer's instructions, shall harden and set to a condition satisfactory for dental use.

3.1.1 Powder. The powder shall be essentially zinc oxide. Modifiers may be added provided cement meets physical properties specified herein. The powder shall be free of extraneous materials such as dirt and lint. The pigment shall be uniformly dispersed throughout the powder.

3.1.2 Liquid. The liquid shall be essentially aluminum phosphate and phosphoric acid. Metallic salts may be added as buffers to reduce the rate of reactivity of the liquid with the powder. The liquid shall be water-clear and there shall be no deposits or sediment on the inside of the container.

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3.1.2.1 Volume. The liquid shall be supplied in an amount, at least, 20% in excess of that necessary to combine with the total amount of powder in a container of full portion size when mixed to testing consistency as determined in 4.4.2.

3.2 Spatulation. Cements when spatulated in the usual manner shall not form lumps or granules, or evolve gas.

3.3 Shades. Shades for the set cement shall be as follows: Shade No. 1 shall be light gray; Shade No. 2 shall be light yellow.

3.4 Physical properties. The zinc phosphate cement shall be fine grain type and shall conform to the physical properties listed in table I, and as herein specified, when tested as specified in section 4.

* Table I. Physical Properties

Testing consistency of mix, disk diameter	Time of setting at 37°C. (99°F.)		Minimum compressive strength after 24 hours	Maximum film thickness	Maximum solubility and disintegration after 24 hours	Maximum arsenic content
	minimum	maximum				
30 ± 0.5 millimeters	5 minutes	9 minutes	770 kg./cm.^{2} (10,949 lb/in^2)	0.025 mm. (0.0010 inch)	0.20% by weight	0.0002% by weight (1:500,000)

Cement shall be suitable for use in the placement of gold inlays, crowns, and all fixed partial prosthetic devices.

3.5 Manufacturer's directions. Adequate and accurate instructions for proportioning and manipulation shall accompany each unit package. These instructions shall include the required temperature of the slab, the powder-liquid ratio, in terms of grams of powder (± 5 percent) to 0.5 milliliter (ml.) of liquid, the rate of incorporating the powder and the time of mixing. In addition, two copies of the instructions shall be forwarded to the contracting officer for information purposes at time of delivery.

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3.6 Lot or control number. Each container of powder and each container of liquid shall be marked with a serial number or a combination of letters and numbers which shall refer to the manufacturer's records for that particular lot or batch of cement powder or liquid.

3.7 Certification. At time of award, suppliers shall furnish a dated certificate and data showing that the brand of **zinc phosphate cement** to be furnished is the same in quality and brand as that appearing on the current list of certified materials for the American Dental Association Specification No. 8 for Dental Zinc Phosphate Cement for Type I, Fine Grain. Compliance with this certification requirement does not absolve the supplier from complete compliance with the requirements of this specification.

3.8 Compatibility. The immediate containers and all components shall be compatible, and shall not interact physically or chemically with the zinc phosphate cement.

3.9 Workmanship. Workmanship shall be first class throughout. Zinc phosphate cement shall be free from defects which may detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II and IV (see 4.2.1 and 4.3) and table III (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels(AQL's) shall be as indicated in table II. Unit of product for examination purposes shall be unit of issue (package).

Table II. Inspection and acceptable quality levels

Type defect	Inspection level	AQL (Percent defective)
Major	S-2	2.5
Minor	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted as indicated in table III. The unit of product for test purposes shall be one package (unit of issue) of dental zinc phosphate cement.

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Table III. Tests

Characteristic	Requirement	Test procedure	Unit of product to be tested per lot or batch number	Number of determinations per unit of product	Total number of determinations
Testing consistency	3.4	4.4.2	1	3 1/	3
Setting time	3.4	4.4.3	1	3 2/	3
Ultimate compressive strength	3.4	4.4.4	1	5 1/	5
Film thickness	3.4	4.4.5	1	3 1/	3
Solubility and disintegration	3.4	4.4.6	2	1 1/	2
Arsenic content	3.4	4.4.7	1	1 2/	1

1/ A lot shall be rejected if the average of the determinations per unit of product fails the test.

2/ A lot shall be rejected if any of the determinations per unit of product fails the test.

4.3 Examination. Zinc phosphate cement shall be examined for defects including, but not limited to, those listed in table IV.

Table IV. Classification of defects

Category	Defect
Major:	
101	All components not present in package.

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Table IV. Classification of defects (cont'd.)

Category	Defect
Major:	
102 1/	Powder not free of extraneous materials such as dirt or lint.
103 1/	Pigment not uniformly dispersed throughout the powder.
104	Liquid not water-clear; deposits or sediment present.
105 1/	Quantities of liquid and powder not as specified.
Minor	
201	Manufacturer's directions not present, not accurate or complete.
202	Lot or control number not present on each container of powder and each container of liquid.
203	Packaging and packing not as specified.
204	Marking not as specified.

1/ In process examination may be used in examination for these defects.

4.4 Tests

4.4.1 Preparation of test specimens. The preparation of all test specimens shall be conducted at 23 ± 2° Centigrade (C.) and 50 ± 10 percent relative humidity. The powder-liquid ratio shall be determined by the consistency test. The following mixing technic shall be employed in the preparation of all test specimens. The polished glass slab shall be approximately 150 millimeters (mm.) long and 75 mm. wide. The spatula (made of material not corroded by the cement) and the slab shall be clean and free from hardened particles of cement. The mixing time shall be 1-1/2 minutes.

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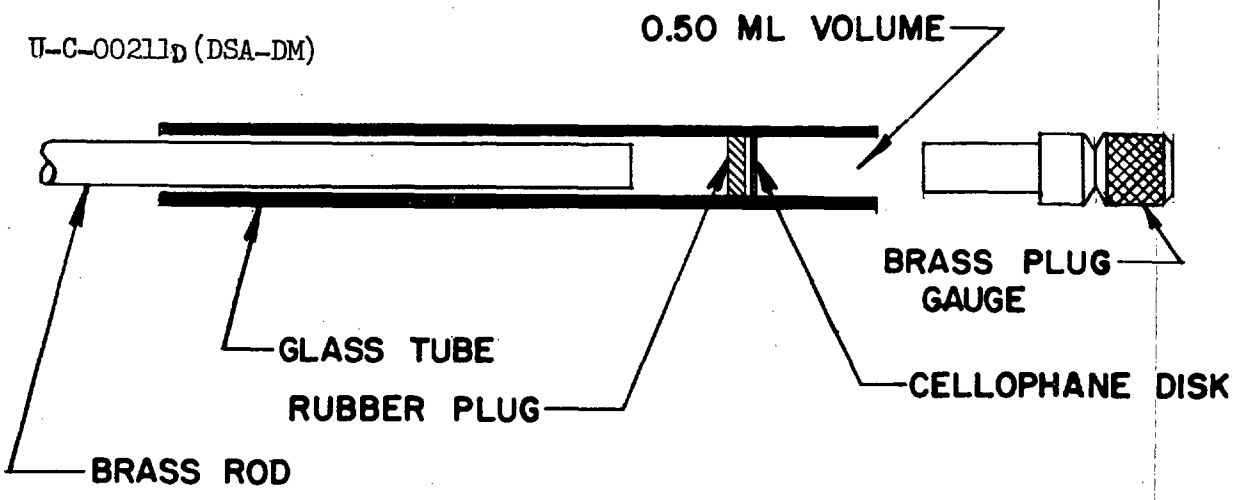
The incorporation of the powder shall be accomplished at the following rate:

Proportion of total amount of powder	Time of incorporation in seconds
1/16	10
1/16	10
1/8	10
1/4	15
1/4	15
1/4	30
	90

At least one half of the top surface of the slab shall be used. No particles of powder or any unused liquid shall remain on the slab when the mixing is completed. A linear and not a rotary motion of the spatula shall be used, with the edge of the spatula sweeping approximately one half of the mixing area of the slab on each stroke.

4.4.2 Testing consistency. The type of apparatus for measuring consistency shall be essentially that shown (Fig. 1). This apparatus consists of two flat glass plates, a weight, a gauge plug and a glass tube (inside diameter approximately 6.5 mm.) which will deliver 0.50 ml. of mixed cement. The combined weight of the top plate and the weight shall be 120 grams (gm.). Trial amounts of powder shall be mixed with 0.50 $\pm$ 0.02 ml. of liquid. Then 0.50 $\pm$ 0.02 ml. of each mix shall be delivered from the glass tube onto a flat glass plate. Three minutes after the mixing is started, the top glass plate, weighing approximately 20 gm., and the necessary additional weight required to total 120 gm. shall be carefully lowered onto the soft cement. Trials shall be made until the average of the major and minor diameters of the slumped mass of cement is 30 $\pm$ 0.5 mm. 10 minutes after starting the mix. The average weight of powder used in three such determinations rounded off to the nearest 0.05 gm. combined with 0.50 ml. of liquid, shall be by definition the testing consistency or amount of powder necessary to produce a mix of standard consistency.

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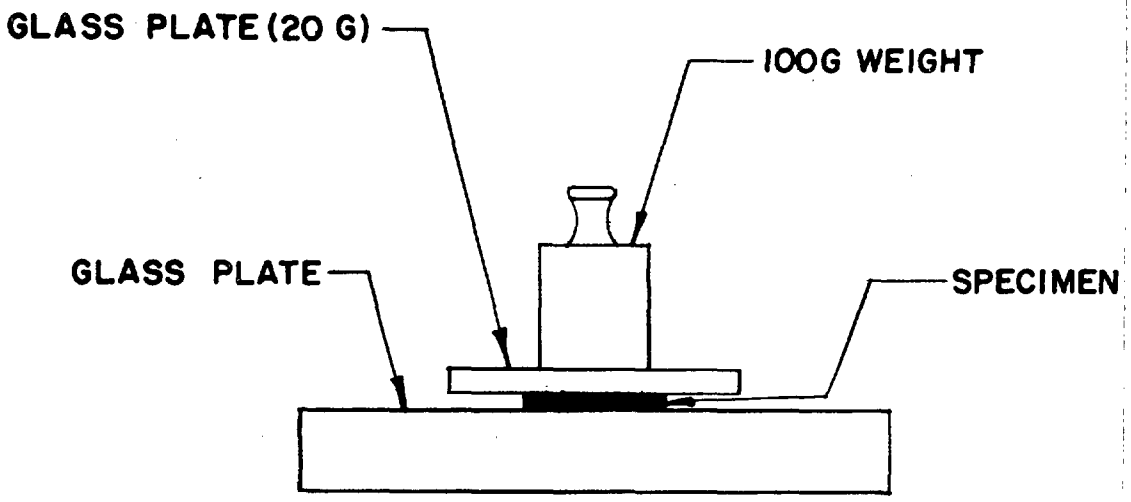


FIGURE 1, APPARATUS FOR CONSISTENCY TESTS

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4.4.3 Time of setting. A type of apparatus found suitable for determining the time of setting is shown (Fig. 2). A brass ring held at room temperature and approximately 4.8 mm. high and 11.1 mm. outside diameter and 9.5 mm. internal diameter shall be placed on a flat glass plate and filled with cement of standard consistency. Three minutes after starting the mix, the specimen shall be transferred to an atmosphere of 100 percent relative humidity at 37 ± 2°C. Three and a half minutes after beginning the mix, a standard Gillmore needle, weighing 453.6 gm. and having an end 1.06 mm. in diameter, shall be carefully lowered vertically onto the horizontal surface of the cement. This shall be repeated at 30-second intervals. The time of setting is the number of minutes elapsed from the starting of the mix to the time when the needle fails to make a perceptible circle on the surface of the specimen when allowed to rest thereon under its own weight. The setting time shall be reported to the nearest minute.

4.4.4 Compressive strength. The test specimens shall be cylinders 12 mm. high and 6 mm. in diameter. The ends of the specimens shall be flat and smooth and shall be parallel to each other and at right angles to the long axis of the cylinder. An apparatus found convenient for forming these test cylinders is shown (Fig. 3). A cylindrical mold 12 mm. high and 6 mm. in diameter shall be placed on a flat glass plate and slightly overfilled with cement of standard consistency within 3 minutes after commencing the mixing. A second flat glass plate shall be pressed on top of the mold. The mold and plates shall be held firmly together with a small C clamp. All apparatus shall be at room temperature. The molds shall be made of hard rubber, glass, stainless steel or any other substance which will not react with the cement. The molds may be coated with a 3 percent solution of a microcrystalline wax (melting point 86-91°C.) in benzene. Three minutes after starting the mix, the mold and clamp shall be transferred to an atmosphere of 100 percent relative humidity at 37 ± 2°C. One hour later the ends of the cylinder shall be surfaced plane at right angles to the axis. The ends of the specimens may be ground flat by the use of a small amount of 200-mesh silicon carbide powder and water. The molds containing the specimens shall be drawn back and forth across a glass plate coated with the abrasive and water. They shall be rotated about one-fourth turn every few strokes. The test specimens shall be kept wet during the grinding. After surfacing, the specimens may be removed from the mold by a screw jack (Fig. 3) and shall be immersed in distilled water at 37 ± 2°C.

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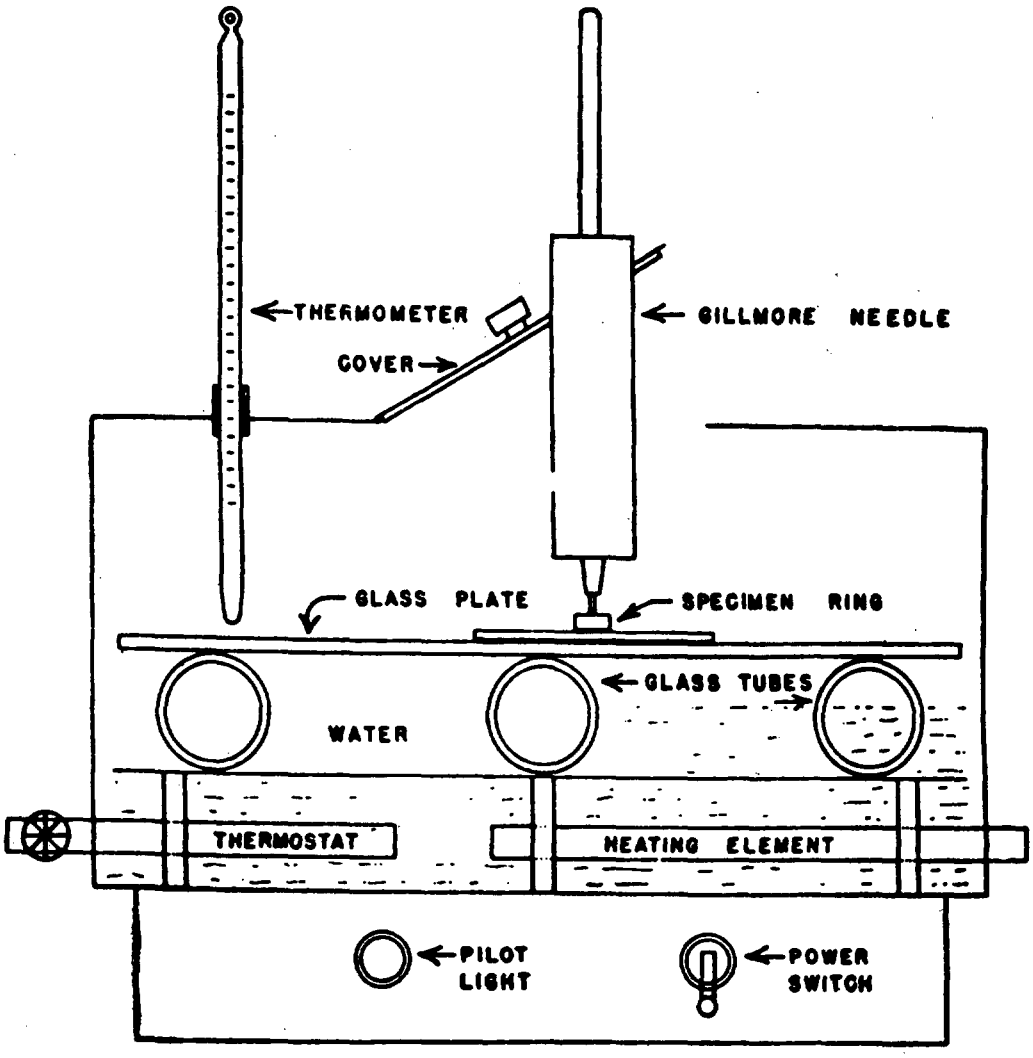
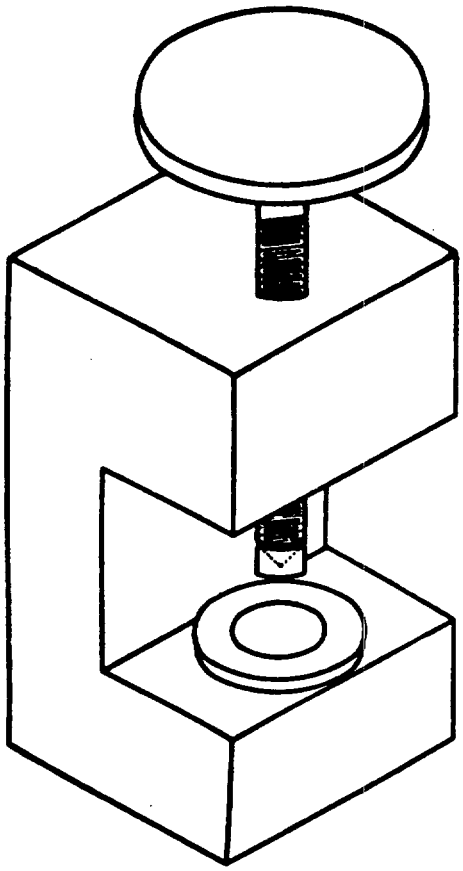


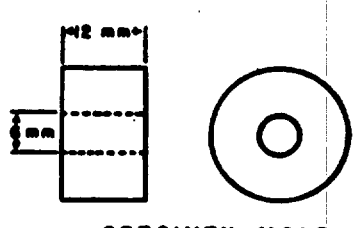
FIGURE 2.—*Apparatus for determining setting time at controlled temperature and humidity.*

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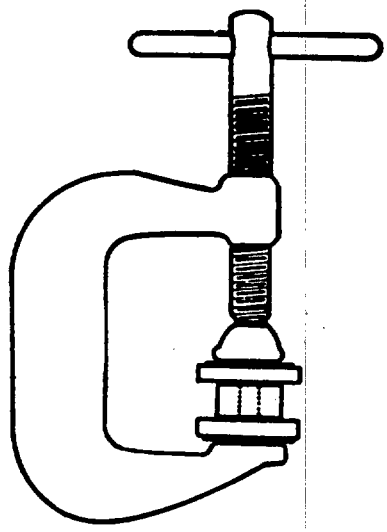
APPARATUS USED IN FORMING ULTIMATE COMPRESSIVE STRENGTH SPECIMENS



SCREW JACK FOR EJECTING SPECIMENS FROM MOLDS



SPECIMEN MOLD



MOLD IN CLAMP

FIGURE 8.—*Apparatus used in forming ultimate compressive strength specimens.*

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The time lapse between the starting of the mixing and the crushing shall be 24 hours. A small piece of blotting paper (approximately 0.5 mm. thick) wet with water shall be inserted between the ends of the specimen and the platens of the testing machine. The specimens shall be loaded at the rate of 225 kilograms (kg.) a minute and shall be kept wet during the test. The value for compressive strength shall be reported as the average of three or more from a lot of five specimens and shall be rounded off to the nearest 10 kg. per square centimeter. If the values for individual specimens fall more than 15 percent below the average of the five, they shall be discarded and the average of the remaining specimens shall be reported. In case more than two of the specimens are eliminated, the test shall be repeated.

4.4.5 Film thickness. A portion of a mix of standard consistency shall be placed between two flat square or round glass plates of uniform thickness. The surface areas of the plates between which the cement is spread shall be approximately 2 square centimeters. Three minutes after the mix is started, a load of 15 kg. shall be applied vertically on the top plate. Ten minutes after the mix is started, the thickness of the two plates with the cement film between them shall be determined. The difference in the thickness of the plates with and without the cement film shall be considered as the film thickness. An average of three tests shall be reported to the nearest 5 microns.

4.4.6 Solubility and disintegration. One-half milliliter of cement of standard consistency shall be placed in a split stainless-steel ring (inside diameter 20 mm. and 1.5 mm. thick) which is sitting on a thin polyethylene sheet backed by a flat plate. Another flat plate faced with a sheet of thin polyethylene shall be used to press the cement into the ring. A piece of fine platinum or corrosion-resistant wire shall be placed in the soft cement as the specimens are formed to provide a convenient means of holding the specimens. Three minutes after the mix is started, the plates and cement shall be placed for 1 hour in an atmosphere having a relative humidity of 100 percent at 37 $\pm$ 2°C. After 1 hour, the specimens shall be withdrawn from the oven and two specimens shall be placed in each tared weighing bottle and weighed (Fig. 4). The combined weight of the two specimens and the weighing bottle, less the weight of weighing bottle and the platinum wire, shall be taken as the weight of the specimens of cement. The specimens shall immediately be submerged by pouring 50 ml. of distilled water into the weighing bottle, which shall be stored for 23 hours at

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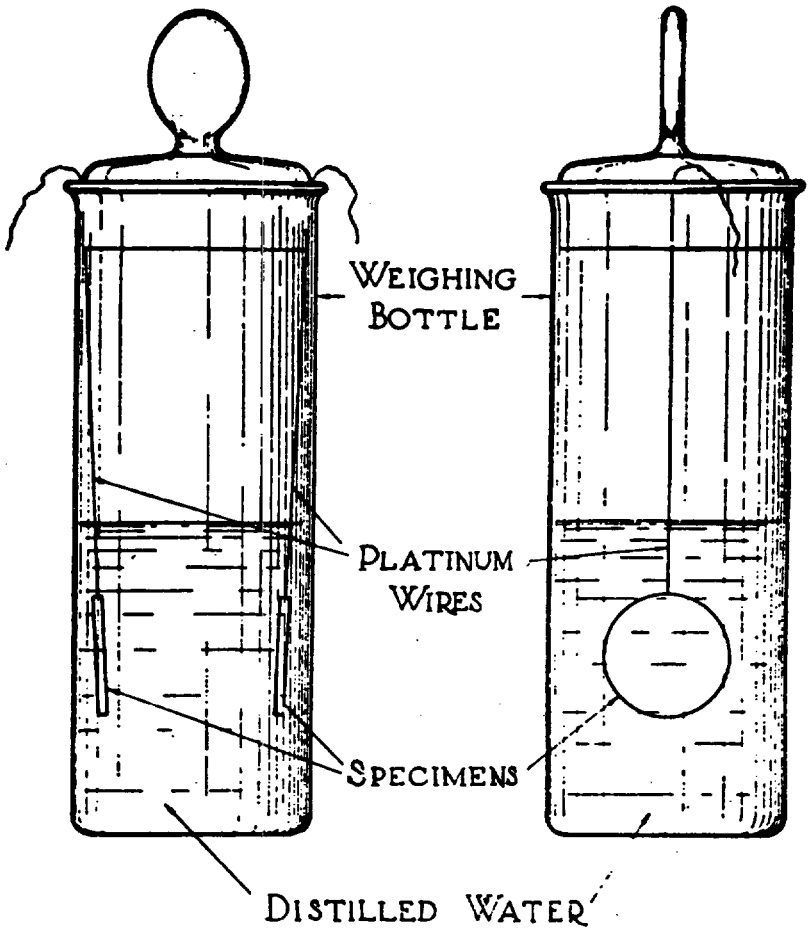


FIGURE 4.—*Weighing bottle containing solubility specimens*

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37 ± 2°C. The specimens shall then be removed from the water. There shall be no evidence of crystal growth on the surface of the specimens. The water shall be evaporated from the weighing bottle at a temperature just below 100°C. The weighing bottle shall then be dried at 150 ± 2°C to constant weight. After cooling to room temperature in a desiccator, containing thoroughly dry anhydrous calcium sulfate ($CaSO_4$) or silica gel (freshly dried at 130°C.), the weighing bottle and contents shall be weighed with a precision of 0.2 mg. This cycle of heating the weighing bottle 150 ± 2°C., cooling over a desiccant and reweighing shall be repeated until the weight loss of each bottle is not more than 0.5 milligram (mg.) in any 24-hour period. The difference between the final weight of the weighing bottle and its initial weight shall be the amount of disintegration. The gain in weight divided by the weight of the specimens times 100 gives the percentage of disintegration. The average of duplicate tests (two weighing bottles containing two specimens each) shall be reported to the nearest 0.01 percent.

4.4.7 Arsenic content. One gram of hardened cement, obtained from a specimen 24 hours old, that has been stored in a dry air-tight container shall be powdered to pass a no. 200 sieve. The powdered sample shall be digested in 100 ml. of distilled water on a steam bath for 1 hour. The filtrate shall be used in the test for water-soluble arsenic. The test for arsenic shall conform to that designated in Reagent Chemicals, American Chemical Society Specifications (2.1).

5. Preparation For Delivery.

5.1 Packaging. Packaging shall be Level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One package, consisting of one ounce of cement powder, and a sufficient quantity of liquid to provide an amount of twenty percent in excess of that necessary to combine with the specified amount of powder when mixed to standard testing consistency, constitutes one unit of issue.

5.1.2 Level A.

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* 5.1.2.1 Unit package. The portions of cement powder and liquid constituting one unit shall be individually packaged in suitable glass bottles. The powder bottle shall be closed with a continuous thread plastic screw cap having a suitable inside liner. The liquid bottle shall be tightly sealed with a suitable cork stopper in the neck of the bottle and covered with a continuous thread plastic screw cap having a cut-out for insertion of the rubber cap. The plastic cap shall have a suitable removable cap liner which can be discarded upon removal of the cork stopper and insertion of the dropper. A glass, liquid dropper with a suitable molded rubber cap to fit and seal the liquid bottle and screw cap, and a corkscrew shall be included in each unit package. The plastic caps of both the powder and liquid bottles shall have a tight viscose seal covering the exposed cap-bottle junction. There shall be no interaction between the immediate container (including rubber cap) and the liquid and powder of the cement. As an alternate to the cork stopper closure, the liquid bottle shall be tightly sealed with a suitable removable plastic disc seal, held securely by a continuous thread plastic screw cap having a cut-out for insertion of the dropper. Requirements for corkscrew shall not apply to this alternate. The bottles together with required instructions shall be packaged in a box of appropriate size constructed in accordance with PPP-B-676, except commercial colors will be acceptable. The box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Closure shall be secured to prevent accidental opening.

* 5.1.2.2 Intermediate package. Six unit packages shall be packaged in a snug fitting box constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be secured to prevent accidental opening under normal handling.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C, as specified (see 6.1).

* 5.2.1 Level A. Sixteen intermediate packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification. In addition, fiberboard boxes shall conform to the special requirements as specified in PPP-B-636.

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* 5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

* 5.2.2 Level B. Sixteen intermediate packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. In addition, fiberboard boxes shall conform to the special requirements as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Shade required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

* 6.2 Specification changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, or deletions) from the previous issue were made. The marking was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the previous issue.

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* 6.3 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

Stock No.	Item Identification
6520-00-514-2350	CEMENT, ZINC PHOSPHATE, DENTAL, Light Gray, 1 oz.
6520-00-514-2450	CEMENT, ZINC PHOSPHATE, DENTAL, Light Yellow, 1 oz.

6.4 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1374

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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